

Chapter 5

Cooling of water

5.1 Objectives

1. To verify newton's law of cooling for water
2. To determine the cooling constant of water using the graphical method

5.2 Tools

1. Three (3) beakers (600 *ml*, 250 *ml*, and 120 *ml*).
2. Three (3) glass thermometers, with a minimum scale of 1°C.
3. A thermos filled with hot water.

5.3 Theoretical Foundation

Newton's law of cooling state that the rate of decrease in the temperature of a substance is proportional to the difference between the temperature of the substance and the ambient (surrounding) temperature. In a mathematical equation:

$$\frac{dT}{dt} = \alpha(T - T_0) \quad (5.1)$$

with α is a constant, T is the temperature of the substance at time t , and T_0 is the ambient temperature. Equation (5.1) may be expressed as:

$$\frac{dT}{T - T_0} = \alpha \cdot dt \quad (5.2)$$

Equation (5.2) is integrated:

$$\int_{T_a}^T \frac{dT}{T - T_0} = \int_0^t \alpha \cdot dt \quad (5.3)$$

The solution to equation (5.3) is

$$\ln(T - T_0) \Big|_{T_a}^T = \alpha t \quad (5.4)$$

$$\ln(T - T_0) = \ln(T_a - T_0) + \alpha t \quad (5.5)$$

$$\ln\left(\frac{T - T_0}{T_a - T_0}\right) = \alpha t \quad (5.6)$$

$$T - T_0 = (T_a - T_0)e^{\alpha t} \quad (5.7)$$

$$T = T_0 + (T_a - T_0)e^{\alpha t} \quad (5.8)$$

with T_a being the initial temperature of the substance.

5.4 Experimental Methods

The first term on the right-hand side of equation (5.5) is a constant, therefore if a graph is plotted of $\ln(T - T_0)$ versus t , it will yield a straight line with slope of α and an intercept of $\ln(T_a - T_0)$. By recording the water temperature T at each time t , then the relationship between $\ln(T - T_0)$ and t can be obtained, which can then be used to determine the cooling rate constant α .

5.5 Procedure

1. Record the room temperature.
2. Fill the three beakers (600 ml, 250 ml, and 120 ml) to full capacity with hot water.

3. Insert a thermometer into each beaker. Wait until the mercury or liquid column in the thermometer stops rising.
4. Record the initial temperature of the water in each beaker.
5. Measure and record the temperature of each beaker at 1-minute intervals.
6. Create a table of values for $T - T_0$, then calculate the natural logarithm $\ln(T - T_0)$ for each time point.
7. Plot a graph of $\ln(T - T_0)$ versus t .
8. From that graph, determine the value of α .